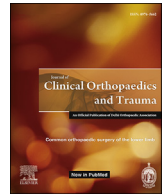




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Allogenic umbilical cord blood-derived mesenchymal stem cells implantation for the treatment of juvenile osteochondritis dissecans of the knee

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ABSTRACT

Background: Osteochondritis dissecans (OCD) is a pathologic condition accompanied by the gradual destruction of subchondral bone and defects in the overlying articular cartilage.

This case series reports the results of allogenic human umbilical cord blood-derived mesenchymal stem cell (hUCB-MSC) implantation for the treatment of osteochondral defect in two cases of juvenile osteochondritis dissecans.

Case presentation: Two patients with osteochondral defect of the knee recovered from the disease enough to begin major exercise 1 year after hUCB-MSCs implantation. The IKDC, VAS, and Tegner score of the two patients showed an excellent improvement and concurrent arthroscopy was performed; cartilage regeneration of ICRS grade 1 similar to normal was observed. The modified two-dimensional MOCART scores increased in both cases over time.

Conclusion: This is the first case series detailing the results of treating juvenile OCD lesions using hUCB-MSCs. This could be an option for treating juvenile OCD.

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1. Case report

The outcome of allogenic human umbilical cord derived mesenchymal stem cells(hUCB-MSCs) therapy in two cases of juvenile osteochondritis dissecans-a case report.

2. Background

Osteochondritis dissecans (OCD) is a pathologic condition accompanied by the gradual destruction of subchondral bone and defects in the overlying articular cartilage. If not treated, early osteoarthritis(OA) develops and continuous pain and disability occurs.¹

Drilling, internal fixation, and microfracture are conducted in

cases of OCD which are unstable or unresponsiveness to conservative treatment^{2,3,4}. Autologous chondrocyte implantation (ACI) or OAT(Osteochondral Autograft transplantation) is also used because cartilage regeneration is difficult with a bone marrow only stimulating technique, such as microfracturing^{5,6,7,8}.

However, ACI is costly and time consuming due to the need for multiple surgeries^{5,6}. OAT have limitations due to the amount cartilage that needs to be gathered from a non-weight bearing area.⁷

Recently, one method introduced mesenchymal stem cells(MSCs) implantation for cartilage defects using human umbilical cords.⁹ This case reports are the results of osteochondral defects in two juvenile OCD patients using allogenic human umbilical cord blood derived mesenchymal stem cells(hUCB-MSCs).

2.1. Preparation of therapeutic mesenchymal stem cells

We used CARTISTEM[®], a medicinal product for the treatment of knee osteoarthritis in this study. CARTISTEM for cartilage repair

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with hUCB-MSCs was produced by MEDIPOST (Seongnam, Gyeonggi-do, South Korea), and the therapeutic use of it for cartilage repair was approved by Ministry of Food and Drug Safety in January 2012.

After informed consent was obtained, MSCs were isolated from the human umbilical cord blood and expanded by repeated subcultures at MEDIPOST. hUCB-MSCs at passage 6 were used as a stem cell drug consisting of mixed formulations with hyaluronate hydrogel for the purpose of treating knee chondral lesions. Prior to implantation, hUCB-MSCs and hyaluronate hydrogel were mixed according to the manufacturer's instructions during surgery (Fig. 1a–d). The therapeutic dosage of CARTISTEM[®] was 500 $\mu\text{l}/\text{cm}^2$ for the defect area with a cell concentration of 0.5×10^7 cells per milliliter^{9,10}.

3. Case presentation

3.1. Case 1

The first patient is a 17-year-old high school wrestler with Lt. Knee pain. He had a limping gait due to knee pain. The patient was 193cm tall and weighed 137kg at the time of diagnosis. OCD at the left knee lateral femoral condyle were evaluated through x-ray and magnetic resonance imaging (MRI) on the first visit. No symptom improvement was observed after 6 months of conservative treatment, such as medication, physiotherapy, and non-weight bearing walking.

A 38 × 28mm bone fragment was noted at lateral side of left knee intraarticular space in the fat suppressed proton-weighted

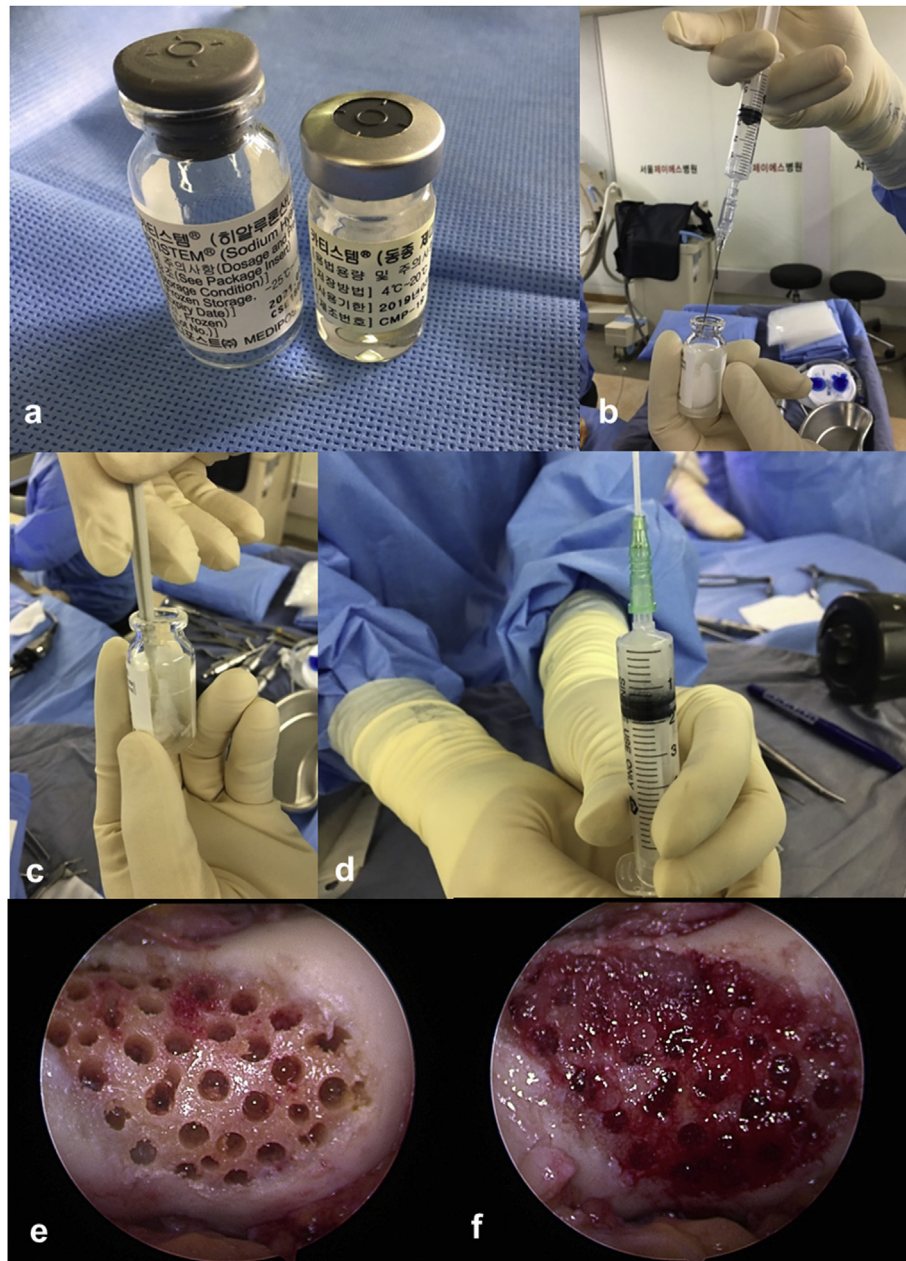


Fig. 1. a) The left vial contains hyaluronic acid and right vial contains hUCB-MSCs. b) Aspirated hUCB-MSCs are injected into the hyaluronic acid vial. c) We mix the hUCB-MSCs with hyaluronic acid. d) The mixture of hUCB-MSCs and hyaluronic acid is put in a syringe. e) After making subchondral holes. f) After injection of the mixture into subchondral holes.

coronal and sagittal images and high signal intensity line was also observed between the bone fragment and the lateral femoral condyle. Several subchondral cysts were also found in the lateral femoral condyle (Fig. 2). Surgical treatment was considered due to a lack of response to conservative treatment, as well as unstable lesions in the MRI^{11,12,13}. The surgical method was tried with cartilage regeneration using hUCB-MSCs after the large bone fragment was removed.

Arthroscopy was conducted through the anteromedial and anterolateral portal. Non-vitalized bone and cartilage fragments were removed from the lateral femoral condyle and unstable cartilage was also removed up to the border of normal tissue. Trimming was conducted on irregular surfaces of subchondral bone using an arthroscopic burr. After arthroscopic procedure, a skin incision was performed. The cartilage defect site was exposed via a 4cm longitudinal lateral arthrotomy, 5mm laterally apart from the patella. Multiple 4mm holes were made in the subchondral bone using 4mm drill bit and an implant mixture of hyaluronic acid-based scaffold and hUCB-MSCs (Fig. 1e and f). The patient was required to rest wearing a knee brace until 3 days after surgery. After 4 days the patient was permitted to walk in a non-weight bearing state.

For 8 weeks after surgery, the patient wore a knee brace and performed non-weight bearing crutch walking. From 8 weeks to 12 weeks, partial weight bearing crutch walking is performed. Starting 4 days after surgery, range of motion with a continuous passive motion machine, quadriceps strengthening exercises, and ankle pump exercises were continuously performed. At 6, 12, and 33 months after surgery, follow-up with MRI was performed. After 35

months, the patient felt pain during knee range of motion (ROM) exercises and a loose body was observed through arthroscopy. The patient felt better after the loose body was removed. The hUCB-MSCs implantation site was difficult to distinguish from normal cartilage, International Cartilage Repair Society (ICRS) grade 1. The strength was similar to normal cartilage when the orthopedic surgeon pressed on the newly formed cartilage using a probe (Fig. 2). A biopsy of the regenerated cartilage was not performed because the patient did not agree to it.

International Knee Documentation Committee (IKDC) scores for pre-op, postop 6 months, 1 year, and 2 years are 17.2, 57.5, 83.9, and 98.9, respectively, increasing as time goes on. Pain seemed to subside with visual analogue scale (VAS) scores at 10, 3, 1, and 0, respectively. The patient was able to perform intensive training through a 1, 4, 6, and 10 level Tegner score, respectively. The patient was able to perform simple exercises, such as running, 6 months after surgery and began performing major exercises 12 months after surgery (Fig. 4). He won the championship after college. He remained as an athlete without any discomfort.

His modified magnetic resonance observation of cartilage repair tissue (MOCART) score was 25, 40, and 60 in a postop MRI at 6, 12, and 33 months, respectively.¹⁴ The score seems to increase as time goes on.

3.2. Case 2

A 16-year-old male with painful swelling of the Rt. Knee. The patient was 178 cm tall and weighed 68 kg. MRI showed a 27 × 22 mm sized, OCD lesion at the trochlea in the right knee. MRI also

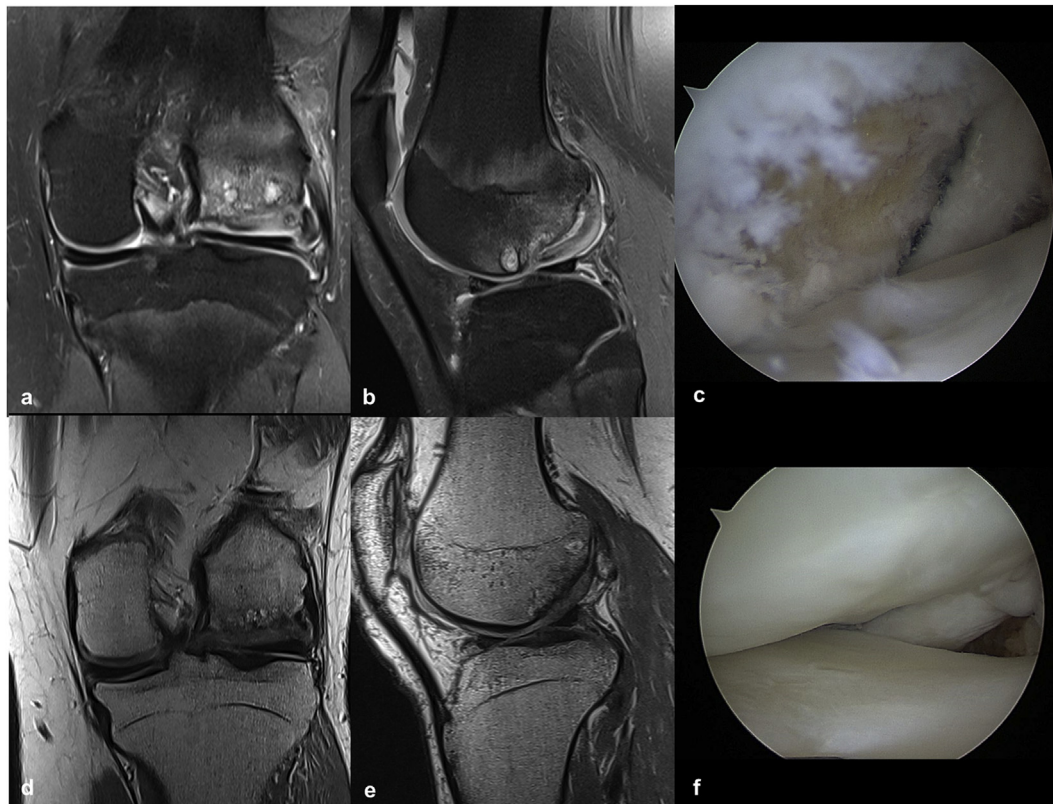


Fig. 2. a, b) Pre-operative coronal and sagittal fat-suppressed proton-weighted MR images of left knee in a 16-year-old male show an osteochondritis dissecans lesion at lateral femoral condyle, a linear high signal intensity line between the bone fragment and lateral femoral condyle as well as multiple subchondral cysts. c) An arthroscopic image demonstrates cartilage defect and exposed irregular subchondral bone surface of the lateral femoral condyle. d, e) Post-operative coronal and sagittal proton-weighted MR images of the left knee at 33 months reveal articular cartilage regeneration (modified 2D MOCART score of 60). Subchondral cysts are still noted but the number and size of cysts are markedly decreased. f) A concurrent arthroscopic image at 35 months depicts cartilage regeneration at hUCB-MSCs implantation site on the lateral femoral condyle.

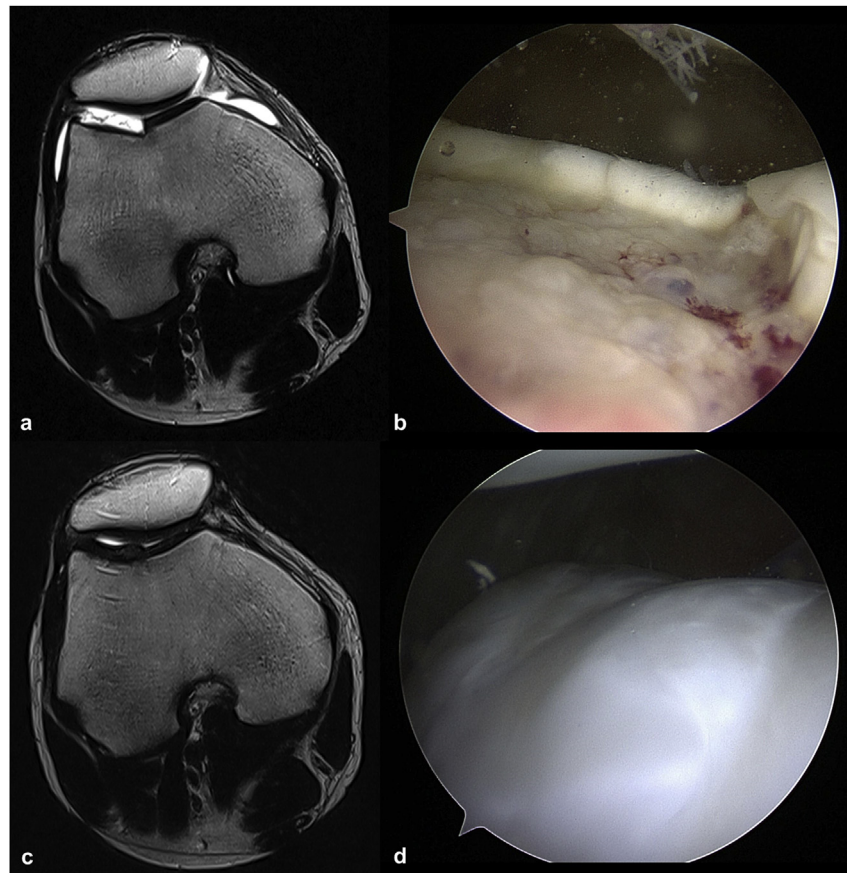


Fig. 3. a) Axial T2-weighted MR image of the right knee in a 16-year-old male reveals a full thickness cartilage defect at the trochlea lateral side. b) An arthroscopic image demonstrates cartilage defect and an exposed irregular subchondral bone surface of the trochlea. c) Axial T2-weighted MR image demonstrates cartilage regeneration at the trochlea osteochondritis dissecans lesion after hUCB-MSCs implantation at 20 months. d) A concurrent arthroscopic image at 20 months depicts cartilage regeneration after hUCB-MSCs implantation at the trochlea.

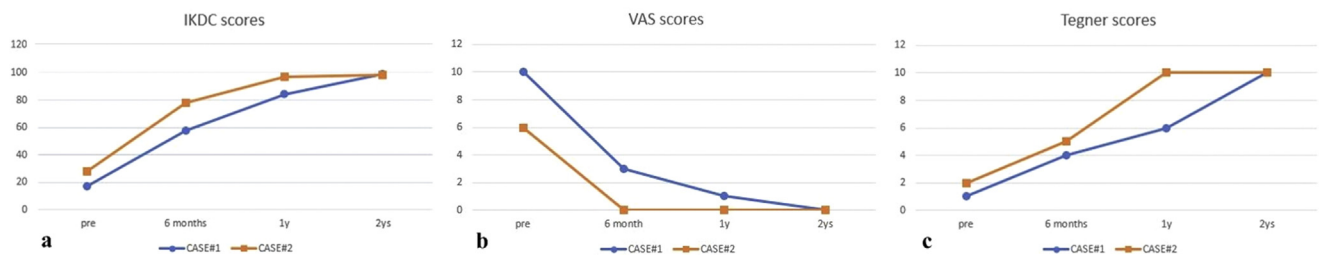


Fig. 4. a) IKDC scores of case 1 and case 2. b) VAS scores of case 1 and case 2. c) Tegner scores of case 1 and case 2.

showed a loose body of cartilage defect at the supra patella bursa. A surgical procedure was considered due to an unstable lesion (Fig. 3). The patient was a high school soccer player at the time of diagnosis. Because the cartilage was grossly defective in size and thickness, a cartilage regeneration technique was considered using hUCB-MSCs in order to get him back to his athletic career quickly. Formal written informed consent was obtained prior to treating the patient through hUCB-MSCs.

Arthroscopy was performed through the anteromedial and anterolateral portal. Non-vitalized bone and cartilage fragments were removed from the trochlea and unstable cartilage was also removed up to the border of normal tissue. Trimming was conducted on irregular surfaces of subchondral bone using an arthroscopic burr. Loose body removal was conducted at the supra patella bursa. After arthroscopic procedure, a skin incision was performed.

The cartilage defect site was exposed via a 4cm longitudinal medial arthrotomy, 5mm medially apart from the patella. Multiple 4mm holes were made in the subchondral bone using 4mm drill bit and an implant mixture of hyaluronic acid-based scaffold and hUCB-MSCs (Fig. 1e and f). The patient was required to bed rest while wearing knee brace for 3 days after surgery and was permitted to walking in a non-weight bearing state at 4 days post-surgery.

Until 8 weeks after surgery, the patient had to wear a knee brace and perform non-weight bearing crutch walking. From 8 to 12 weeks post-surgery, partial weight bearing crutch walking was performed. Starting 4 days after surgery, range of motion with a continuous passive motion machine, quadriceps strengthening, and ankle pump exercises were continuously performed.

Follow-up through MRI was performed 4, 8, and 20 months after surgery. After 20 months, the patient felt left knee joint pain and an

OCD lesion was observed at the trochlea in the left knee MRI. A second arthroscopy of the Rt. Knee was conducted during the Lt. Knee surgery. In arthroscopy, normal cartilage was observed at the implantation site of the hUCB-MSCs, but an elevated surface was observed compared with surrounding cartilage, called ICRS grade 1. Intensity was similar to normal when a surgeon pressed the regenerated cartilage using a probe (Fig. 3). A biopsy of the cartilage was not performed because consent was not obtained from the patient.

The patient was able to perform simple exercises, such as running, 6 months after surgery and began participating in soccer games 12 months after surgery.

Follow-up through MRI was performed 8 months after surgery. The modified 2D MOCART score was 60. Follow-up through MRI was performed 20 months after surgery. The modified 2D MOCART score was 70. The IKDC score was 27.6 before surgery, increasing to 78.2 6 months after surgery, as well as 96.6 and 98.9 after 1 and 2 years after surgery, respectively. The VAS score was 6 before surgery, decreasing to 0, 6 months after surgery then maintaining through 2 years of follow-up. The Tegner activity scale was at a level 2 before surgery, increasing to a level 5, 6 months after surgery, increased to a level 10, 1–2 years after surgery (Fig. 4).

4. Discussion

Efforts were made to cover cartilage loss in cases of accompanying cartilage defect in Juvenile osteochondritis dissecans patients. At first, doctors obtained satisfying results using the microfracturing method, but the results were unsuccessful in long term follow-up. Regenerated cartilage was fibrocartilage, not hyaline cartilage^{3,4}. Alternatively, doctors used ACI or OAT methods that tried to obtain regenerated cartilage similar to hyaline cartilage^{5,6,7,8}. However, ACI is costly and time consuming due to the need for multiple surgeries. OAT has the advantage of repairing cartilage and bone loss quickly, but it has limitations because it has an influence on the trochlea or lateral femoral condyle when harvesting the osteochondral plug.¹³ In order to overcome this limitation, it focuses on the method of regenerating cartilage using differentiation potential to the chondrocyte and paracrine effect of the mesenchymal stem cells.⁹

Recently, one method introduced MSCs implantation for cartilage defects using human umbilical cords.⁹ This case reports are the results of osteochondral defect in two juvenile OCD patients using allogenic human umbilical cord blood derived mesenchymal stem cells (hUCB-MSCs). CARTISTEM[®] that was medicinal product for the treatment of knee OA has been used. CARTISTEM for cartilage repair with hUCB-MSCs was produced by MEDIPOST (Seongnam, Gyeonggi-do, South Korea), and therapeutic use of it for cartilage repair was approved by the Korea food and drug administration in January 2012.⁹ MSCs were isolated from human umbilical cord blood with informed consent and expanded by repeated subcultures at MEDIPOST.

There are several advantages to use hUCB-MSCs among various mesenchymal stem cells. First, hUCB-MSCs are less immunogenic and can avoid host immune surveillance. Because they do not require a close human leukocyte antigen (HLA) match owing to the naive nature of a new-born's immature system. Because of this reason, we can use hUCB-MSCs regardless of gender, allergy, and blood grouping^{10,9}. Second, they showed high expanding capacity compared with bone marrow derived mesenchymal stem cells (BM-MSCs).¹⁵ Third, hUCB-MSCs are 'off-the shelf' product and we can use it whenever we want and as much as we need.¹⁶

Meanwhile, we evaluated the regenerated cartilage through a modified 2D MOCART score. The score seems to be increasing in both cases as time goes on. The score of case 1 was lower than that

of case 2 because pre-operative MRI images of the case 1 showed the large cartilage defect size, the heterogenous signal intensity in subchondral bone, and the variable size subchondral bone cysts. The reason for a lower modified 2D MOCART score of case 1 at 33 months follow-up was considered to be persistent subchondral bone cysts, and an irregular contour of the subchondral lamina found MR images. A modified 2D MOCART score seems to be low in pre-operative and follow-up MRI, if the juvenile OCD lesion is large and the subchondral bone involvement is severe. Although modified 2D MOCAR scores were low, clinical symptoms are much improved in our study.

More study is needed concerning cartilage regeneration using hUCB-MSCs. As far as we know, there are no other case reports about juvenile OCD treatment using hUCB-MSCs. This study is the first case report detailing the results of treating juvenile OCD lesions using hUCB-MSCs; it could be an option for treating juvenile OCD.

Acknowledgements

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Abbreviations

ACI	Autologous chondrocyte implantation
hUCB-MSCS	Allogenic human umbilical cord blood derived mesenchymal stem cells
ICRS	International Cartilage Repair Society
IKDC	International knee documentation committee
MOCART	Magnetic resonance observation of cartilage repair tissue
MRI	Magnetic resonance imaging
OA	Osteoarthritis
OAT	Osteochondral autograft transplantation
OCD	Osteochondritis dissecans
ROM	Range of motion
VAS	Visual analogue scale

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Availability of data and materials

All data generated during this study are included in this published article.

Consent for publication

Consent was obtained from two patients themselves.

Conflicts of interest

The authors declare that they have no competing interests.

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